RMIT Classification: Trusted

ECON1064 – Forecasting and Quantitative Analysis

Assessment 3: Final Assessment

Instructions:

This assignment is to be completed individually. It involves analysing data, estimating forecasting models, carrying out appropriate tests, comparing models on the basis of accuracy measures and interpreting results.

Submission: Via Assignments folder in Canvas.

Marks: The assessment comprises 100 marks and will contribute 40% to your overall grade.

Academic integrity: This is an individual piece of assessment. Submission will be verified via Turnitin for any form of plagiarism. The assessment should contain your own work and you cannot copy or have someone else complete any part of the work for you. By submitting this assessment, you are declaring that you have read, understood and you agree to the content and expectations of the Assessment declaration:
https://www.rmit.edu.au/students/my-course/assessment-results/assessment

Presentation Instructions:

You will submit two files:

1) An R file with all the codes, clearly presented. Your code should run without errors.
2) A Word (.doc or .docx) or PDF (.pdf) file where you will answer the questions in the order that they have been asked. Your document should comply with the following presentation standards:
    a) Typed using a standard professional font type. Font "Arial" size 11 is recommended.
    b) Pages should be numbered.
    c) Label your answer to each question clearly – e.g., Question 1 a.
    d) Graphs and tables should be clearly labelled and presented.
    e) Your work should be well-presented with no spelling, typographical and grammatical errors.
    f) Answer the questions in **a new Word document**. DO NOT copy the questions as this will affect your Turnitin score.

Three files are available to you on Canvas:

1. *Tourist_data.xls*
2. *My_data_SIM.xls*
3. *Final Assessment.R*

The ***Tourist_data.xls*** dataset contains the number of short-term visitors arriving in Australia from selected countries between January 1991 and December 2019. The data is sourced from the Australian Bureau of Statistics (https://www.abs.gov.au/statistics/industry/tourism-and-transport/overseas-arrivals-and-departures-australia/latest-release#data-downloads). Your task is to analyse the data, generate forecasts, conduct relevant tests, perform diagnostic analyses, and produce accuracy measures to compare different models.

R codes (10 marks): A template, ***Final Assessment.R*** is provided. It includes codes for reading the data. You will first locate the Country ID (a unique identifier for each country in tourist_data.xls) that corresponds to your student ID in ***My_data_SIM.xls***. You will then enter this Country ID into the R file to extract data for your assigned country. You will save your R template as ***FamilyName_StudentID.R***. To score well,

1. ensure the code runs smoothly in a single execution.
2. clearly present your code, labelling responses appropriately (e.g., Question 1, Question 2 etc).
3. include comments, where necessary, to document your work and provide clarifications.

Part A (35 marks): The aim in this part of the assignment is to understand the data, perform transformation (if required), and use a simple forecasting method to produce forecasts.

Question 1

Produce appropriate plots in order to become familiar with your data. Make sure you label your axes and plots appropriately. Comment on the plots. What do you see? (50 words per plot). (15 marks)

Question 2

Would transforming your data be useful? If transformation is required, compare two approaches graphically. Choose the best transformation, justifying your choice (100 words). (6 marks)

Question 3

Apply an appropriate benchmark or simple forecasting model to the data, justifying your choice (50 words). (4 marks)

Question 4

Perform a thorough residual analysis. Do the residuals appear to be white noise? (100 words). (6 marks)

Question 5

Generate and plot forecasts and forecast intervals for the next 2 years, also plotting the observed data. You may choose to plot on a shorter period of say 5 last years for clearer visualisation. Discuss your findings (50 words). (4 marks)

Part B (35 marks): The aim in this part of the assignment is to build an ARIMA model and an exponential smoothing (ETS) model and use it to forecast.

Question 6a

Visually inspect your transformed data and decide what differencing is required to achieve stationarity. Analyse using relevant plots at every step, commenting on each plot and justifying your actions. (50 words per plot). (9 marks)

RMIT Classification: Trusted

Question 6b

Estimate an ARIMA model using the auto-ARIMA function in R. Report your results. (3 marks)

Question 7

Estimate an exponential smoothing model using the ETS class of models. Report your results. (3 marks)

Question 8

Perform a thorough residual diagnostics analysis for each of the estimated models (ARIMA and the ETS). Discuss your results. (100 words). (12 marks)

Question 9

Generate and plot together forecasts and forecast intervals for the two models, for the next two years. You may choose to plot on a shorter period of say 5 last years for clearer visualisation. Discuss your findings (100 words). (8 marks)

Part C (20 marks): You have now built three models with your dataset. Next, the aim is to evaluate the three models.

Question 10

Create a training set with your data by leaving two years' worth of observations as the test set. Generate forecasts for the last two years (the period of the test set), from the three models you have estimated in Parts A (simple forecasting model) & B (ARIMA and ETS models). Plot the forecasts (both point forecasts and prediction intervals) together with the observed data and comment on these (100 words). You may choose to plot on a shorter period of say 5 last years for clearer visualisation. (15 marks)

Question 11

Compute the accuracy of your forecasts generated from the three models in a table. Which model does best and why? (50 words) (5 marks)